

Weijia Zhang

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RESEARCH SUMMARY

I study how intelligent systems learn, reason, and adapt as their environments evolve. My research investigates the mechanisms of representation updating and continual learning across biological and artificial systems, and why these mechanisms enable flexible inference and generalization in some settings but produce systematic failures in others.

EDUCATION

Columbia University

Ph.D. in Biomedical Engineering

New York, NY

Expected Spring 2027

Columbia University

M.S. in Biomedical Engineering

New York, NY

2021

Boston University

B.A. in Applied Mathematics, Neuroscience, and Economics

Boston, MA

2020

TECHNICAL SKILLS

Programming & Tools: Python, MATLAB, Bash, Git, Linux, LaTeX

ML Frameworks & Computing: PyTorch, JAX, TensorFlow, Hugging Face, scikit-learn, SLURM/HPC

Research Areas: LLM Evaluation, Neuro-AI, Reinforcement Learning, Continual Learning, Adaptive Reasoning

RESEARCH EXPERIENCE

Columbia University, Center for Theoretical Neuroscience | Biomedical Engineering

Ph.D. Researcher

New York, NY

Sep 2022 – Present

Project: LLM Evaluation and Adaptive Reasoning

May 2026 – Present

Research Advisors: Kimberly L. Stachenfeld (Columbia / Google DeepMind) & Jesse P. Geerts (Callosum)

- Designed controlled multi-turn LLM revaluation benchmarks to investigate how models revise learned associations, arbitrate between cached knowledge and explicit reasoning, and propagate new information to downstream inferences.
- Systematically manipulated task novelty, inference depth, revaluation type, and reasoning strategy to investigate failure modes in knowledge updating, relational reasoning, and non-local inference.
- Built scalable local and API-based evaluation pipelines for open-weight and closed-source models across scales; analyzed behavioral signatures and reasoning traces to characterize successful adaptation and systematic failure modes.

Project: Neural and Computational Mechanisms of Adaptive Learning

Sep 2022 – Present

Research Advisors: Joshua Jacobs (University of Chicago) & Kimberly L. Stachenfeld (Columbia / Google DeepMind)

- Designed controlled revaluation experiments to isolate how humans selectively update learned representations and infer non-local consequences when values or environmental structure change.
- Modeled adaptive learning across reinforcement learning agents, RNNs, and GPT-style Transformers to compare computational strategies for updating learned representations; used representational analyses and attention ablations to characterize mechanisms underlying successful and failed adaptation.
- Linked computational predictions to large-scale human single-neuron recordings to identify representational signatures of adaptive updating; built parallelized analysis pipelines and contributed to SpikeTools and HSNPipeline for scalable electrophysiology analysis.

Columbia University, Laboratory for Intelligent Imaging and Neural Computing

Master's Student Researcher | Advisor: Dr. Paul Sajda

New York, NY

Sep 2020 – Aug 2022

Project: Multimodal Predictive Modeling of Human Team Coordination

- Developed multi-head attention models to predict teammates' future actions from multimodal EEG, pupillometry, speech, and behavioral signals; identified action predictability as a stronger predictor of team performance than synchronization.

McGovern Institute for Brain Research at MIT, MEG Lab

Undergraduate Researcher | Advisor: Dr. Dimitrios Pantazis

Cambridge, MA

Sep 2019 – Sep 2020

Project: Neural Dynamics of Human Visual Processing

- Applied multivariate pattern analysis and SVM classifiers to decode visual representations from MEG recordings; optimized stimulus presentation rates to reduce experimental duration by 30% while preserving decoding performance.

PUBLICATIONS

Peer-Reviewed Publications

Zhang W, Donoghue T, Qasim SE, Jacobs J. *Evaluating Place Cell Detection Methods in Rats and Humans: Implications for Cross-Species Spatial Coding*. PLOS Computational Biology, 22(5), e1013488.

Donoghue T, **Zhang W**, Han CZ, Maesta-Pereira S, Jacobs J. *HSNPipeline: A Processing Pipeline for Human Single-Neuron Studies*. Aperture Neuro, 6(SI 1).

Qin Y, Lee RT, **Zhang W**, Sun X, Sajda P. *Physiologically Informed Predictability of a Teammate's Future Actions Forecasts Team Performance*. iScience, 28(5), 112429 (2025).

Qin Y, **Zhang W**, Lee R, Sun X, Sajda P. *Predictive Power of Pupil Dynamics in a Team-Based Virtual Reality Task*. IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW) (2022).

Manuscript Under Review

Tostaeva G, Belilty K, Qasim S, **Zhang W**, Jacobs J, Ghatan S, Panov F, Schiller D, Saez I. *Human Hippocampal and Orbitofrontal Neurons Track Latent Position in Social Space*. Under review at Nature Neuroscience.

Manuscripts in Preparation

Zhang W^{*}, Lippel S^{*}, Geerts JP, Stachenfeld KL. *Cognitive Arbitration in Large Language Models during Multi-Turn Adaptive Reasoning*.

Zhang W, Donoghue T, Maesta-Pereira S, Tostaeva G, Carranza OA, Lega B, Saez I, Geerts JP, Stachenfeld KL, Jacobs J. *Human Single Neurons and Predictive Models Reveal Representational Mechanisms of Flexible Adaptation*.

Zhao CK, **Zhang W**, Zhang Y, Paninski L, Jacobs J. *Introducing CoMETs: Contrastive Multimodal Embeddings for Time-Series in Neuroscience*.

^{*}Equal contribution.

SELECTED PRESENTATIONS

Invited Talk: Spring Hippocampal Symposium, Verona, Italy (2027; upcoming).

Contributed Talk: Gatsby Tri-Center Meeting, London, UK (2026).

Data Blitz: iNav, Banff, Canada (2026).

Selected Posters: Cosyne (2026, 2025); CCN (2026); CEMS(2026); Society for Neuroscience (2025, 2024, 2023).

TEACHING & PROFESSIONAL SERVICE

Teaching: Teaching Assistant, *Biostatistics for Engineers; AI in Medicine*.

Research Outreach: Ambassador, *Thinking About Thinking* (Apr 2026 – Present).

Conference Service: Organizer, proposed *I Can't Believe It's Not Better (ICBINB)* Workshop for ICLR 2027; Reviewer, ICBINB Workshop, ICLR 2025.

Peer Review: Assisted peer review with Joshua Jacobs for *Science* (2026), *Trends in Cognitive Sciences* (2025), *eLife* (2024), *Nature Human Behaviour* (2023), and *Nature Communications* (2023).